Economic growth and Life expectancy in the world

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After advanced research of the provided data about Economic growth, meaning the quantity or quality of the many goods and services that people produce and Life expectancy throughout the contries in the world, all features were taken into account in order to retrieve meaningful information about the contries and their population.

Imports

In [18]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Loading the data

GDP per capita

In [19]:
```
gdp_df = pd.read_csv("gdp-per-capita-worldbank.csv")
gdp_df
```

Out[19]:

	Entity	Code	Year	GDP per capita, PPP (constant 2021 international $)	World regions according to OWID
0	Afghanistan	AFG	2000	1617.8264	NaN
1	Afghanistan	AFG	2001	1454.1108	NaN
2	Afghanistan	AFG	2002	1774.3087	NaN
3	Afghanistan	AFG	2003	1815.9282	NaN
4	Afghanistan	AFG	2004	1776.9182	NaN
...	...	...	...	...	...
7306	Zimbabwe	ZWE	2020	2987.2683	NaN
7307	Zimbabwe	ZWE	2021	3184.7847	NaN
7308	Zimbabwe	ZWE	2022	3323.1184	NaN
7309	Zimbabwe	ZWE	2023	3442.2488	Africa
7310	Zimbabwe	ZWE	2024	3450.1296	NaN

7311 rows × 5 columns

There are 7311 entries with 5 features, three of them are categorical and the Code and GDP per capita columns have some null entries. In comparison the Entity and Year features are fully completed unlike the World regions feature which contains too many null entries to be used for obtaining information.

In [20]:
```
gdp_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7311 entries, 0 to 7310
Data columns (total 5 columns):
 #   Column                                          Non-Null Count  Dtype  
---  ------                                          --------------  -----  
 0   Entity                                          7311 non-null   object 
 1   Code                                            6876 non-null   object 
 2   Year                                            7311 non-null   int64  
 3   GDP per capita, PPP (constant 2021 international $)  7236 non-null   float64
 4   World regions according to OWID                 272 non-null    object 
dtypes: float64(1), int64(1), object(3)
memory usage: 285.7+ KB
```

Life expectancy

In [21]:
```
life_expectancy_df = pd.read_csv("life-expectancy.csv")
life_expectancy_df
```

Out[21]:

	Entity	Code	Year	Period life expectancy at birth
0	Afghanistan	AFG	1950	28.1563
1	Afghanistan	AFG	1951	28.5836
2	Afghanistan	AFG	1952	29.0138
3	Afghanistan	AFG	1953	29.4521
4	Afghanistan	AFG	1954	29.6975
...	...	...	...	...
21560	Zimbabwe	ZWE	2019	61.0603
21561	Zimbabwe	ZWE	2020	61.5300
21562	Zimbabwe	ZWE	2021	60.1347
21563	Zimbabwe	ZWE	2022	62.3601
21564	Zimbabwe	ZWE	2023	62.7748

21565 rows × 4 columns

Life expectancy dataset contains 2 categorical and 2 numerical features. The important features are fully completed, so they can be used for retrieving accurate information.

In [22]:
```
life_expectancy_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21565 entries, 0 to 21564
Data columns (total 4 columns):
 #   Column                            Non-Null Count  Dtype  
---  ------                            --------------  -----  
 0   Entity                            21565 non-null  object 
 1   Code                              19609 non-null  object 
 2   Year                              21565 non-null  int64  
 3   Period life expectancy at birth  21565 non-null  float64
dtypes: float64(1), int64(1), object(2)
memory usage: 674.0+ KB
```

Before continuing, we will make the GDP and Life expectancy feature names easily readable.

In [23]:
```
gdp_df = gdp_df.rename(columns={'GDP per capita, PPP (constant 2021 internationa
life_expectancy_df = life_expectancy_df.rename(columns={'Period life expectancy
```

Statistical data

Information about Life expectancy and GDP per capita statistical information will be used for further investigation of the dataset as well as answering pressing issues about the countries economies.

In [24]:
```
life_expectancy_df.describe().T
```

Out[24]:

	count	mean	std	min	25%	50%	7
Year	21565.0	1976.992812	38.461606	1543.0000	1962.000	1982.0000	2003.0
Life_expectancy	21565.0	61.942232	12.925912	10.9891	52.703	64.4799	71.9

In [25]:
```
gdp_df.describe().T
```

Out[25]:

	count	mean	std	min	25%	50%
Year	7311.0	2007.349063	10.124581	1990.0000	1999.00000	2008.000
GDP_per_capita	7236.0	21729.221482	23785.629715	510.8228	4372.77095	12710.775

Results

Firstly, a scatter plot will be displayed, using the latest year visible in both datasets and then we will compare the results with a couple of other available years.

In [26]:
```
gdp_years = set(gdp_df['Year'].unique())
life_expectancy_years = set(life_expectancy_df['Year'].unique())
common_years = sorted(list(gdp_years.intersection(life_expectancy_years)))

latest_year = common_years[-1]
latest_year
```

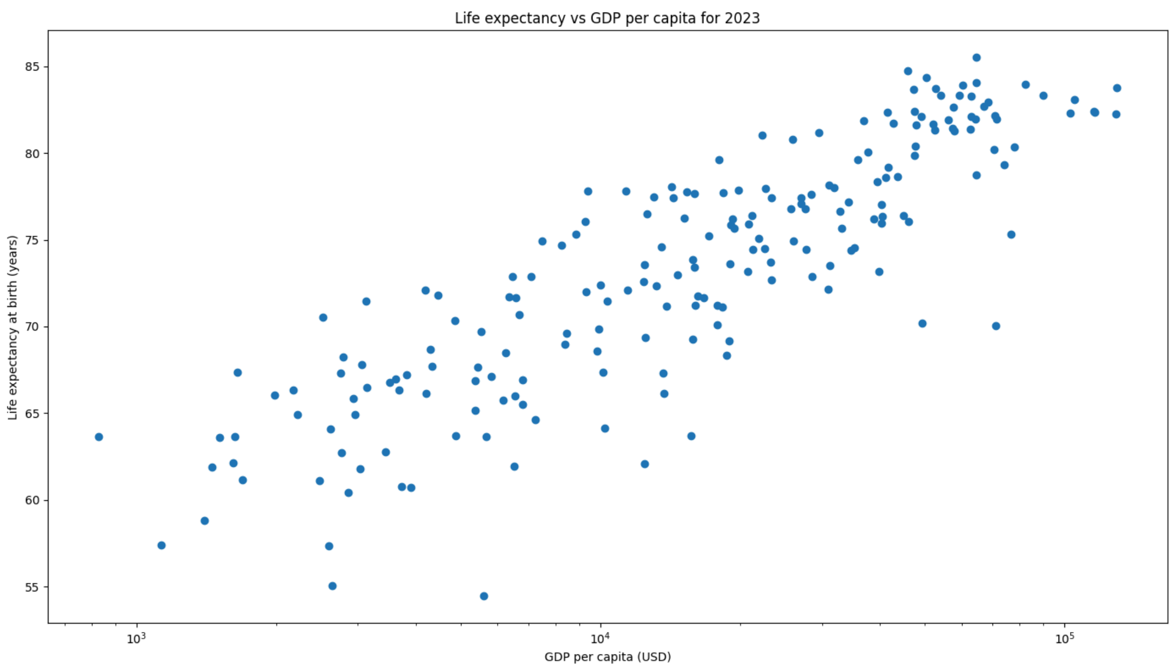
Out[26]: 2023

We will use the latest year for investigating the correlation between the GDP per capita and Life expectancy. We will use the logarithmic scale for GDP per capita in order to decrease the variety and to get a tighter and more informational scatter plot.

The significant features which as we investigated above are the Country, Life expectancy and GDP per capita. These features will provide the initial information. The other features can be excluded because either they are giving the same information or have more than 75% missing data which can change the results.

In [27]:
```
gdp_2023 = gdp_df[gdp_df['Year'] == latest_year][['Entity', 'GDP_per_capita']]
life_expectancy_2023 = life_expectancy_df[life_expectancy_df['Year'] == latest_y

merged_df = pd.merge(gdp_2023, life_expectancy_2023, on='Entity', how='inner')

merged_df = merged_df.dropna(subset=['GDP_per_capita', 'Life_expectancy'])

merged_df = merged_df.dropna(subset=['GDP_per_capita', 'Life_expectancy']).reset
```

In [28]:
```
plt.figure(figsize=(14,8))
plt.scatter(merged_df['GDP_per_capita'], merged_df['Life_expectancy'])
plt.xscale('log')
plt.xlabel(f'GDP per capita (USD)')
plt.ylabel('Life expectancy at birth (years)')
plt.title('Life expectancy vs GDP per capita for 2023')

plt.tight_layout()
plt.show()
```



A strong correlation between the GDP per capita and the Life expectancy of a person can be seen. The more the GDP per capita is, the more is the average life expectancy of a person in the country.

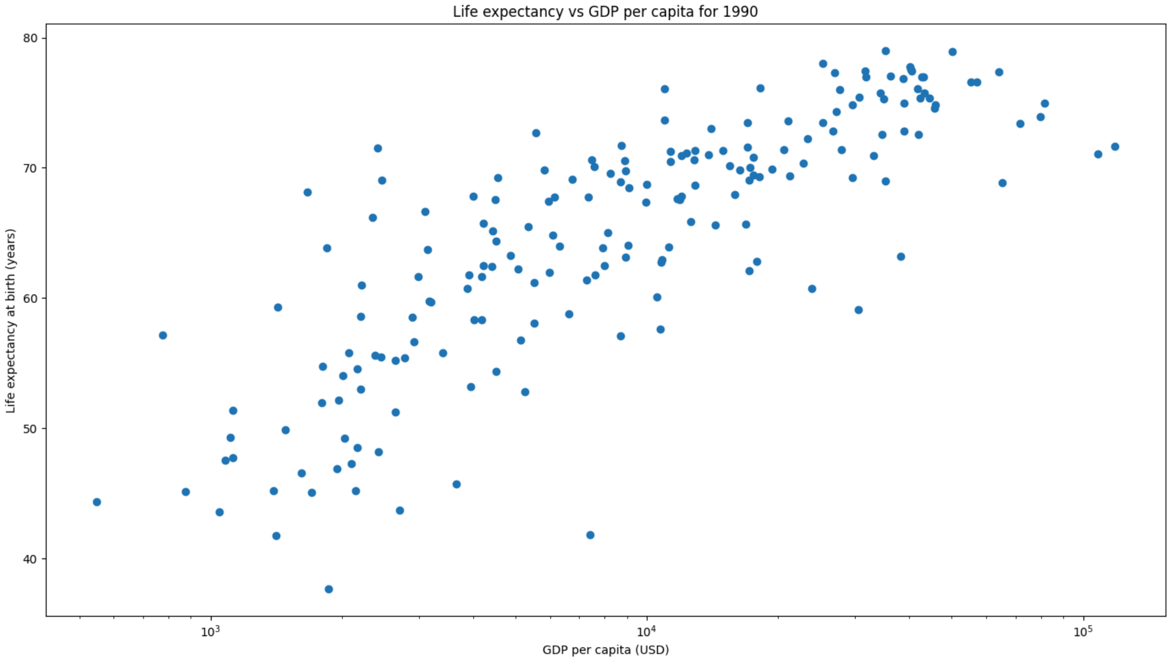
Let's now compare this result to the earliest year visible in both datasets.

In [29]:
```
earliest_year = common_years[0]
earliest_year
```

Out[29]: 1990

In [30]:
```
gdp_1990 = gdp_df[gdp_df['Year'] == earliest_year][['Entity', 'GDP_per_capita']]
life_expectancy_1990 = life_expectancy_df[life_expectancy_df['Year'] == earliest

merged_df_1990 = pd.merge(gdp_1990, life_expectancy_1990, on='Entity', how='inne

merged_df_1990 = merged_df_1990.dropna(subset=['GDP_per_capita', 'Life_expectanc

merged_df_1990 = merged_df_1990.dropna(subset=['GDP_per_capita', 'Life_expectanc
```

In [31]:
```
plt.figure(figsize=(14,8))
plt.scatter(merged_df_1990['GDP_per_capita'], merged_df_1990['Life_expectancy'])
plt.xscale('log')
plt.xlabel(f'GDP per capita (USD)')
plt.ylabel('Life expectancy at birth (years)')
plt.title('Life expectancy vs GDP per capita for 1990')

plt.tight_layout()
plt.show()
```



In conclusion, we can see that the correlation between the two features is also visible in 1990 although there are some more distractions in the data.

Analyzing the results

b. Which countries have a life expectancy higher than one standard deviation above the mean?

In [38]:
```
life_expectancy_mean = merged_df['Life_expectancy'].mean()
life_expectancy_std = merged_df['Life_expectancy'].std()
life_expectancy_requirements = life_expectancy_mean + life_expectancy_std
```

```
high_life_expectancy = merged_df[merged_df['Life_expectancy'] > life_expectancy_
    'Life_expectancy', ascending=False
).reset_index(drop=True)

print(high_life_expectancy[['Entity','Life_expectancy','GDP_per_capita']])
```

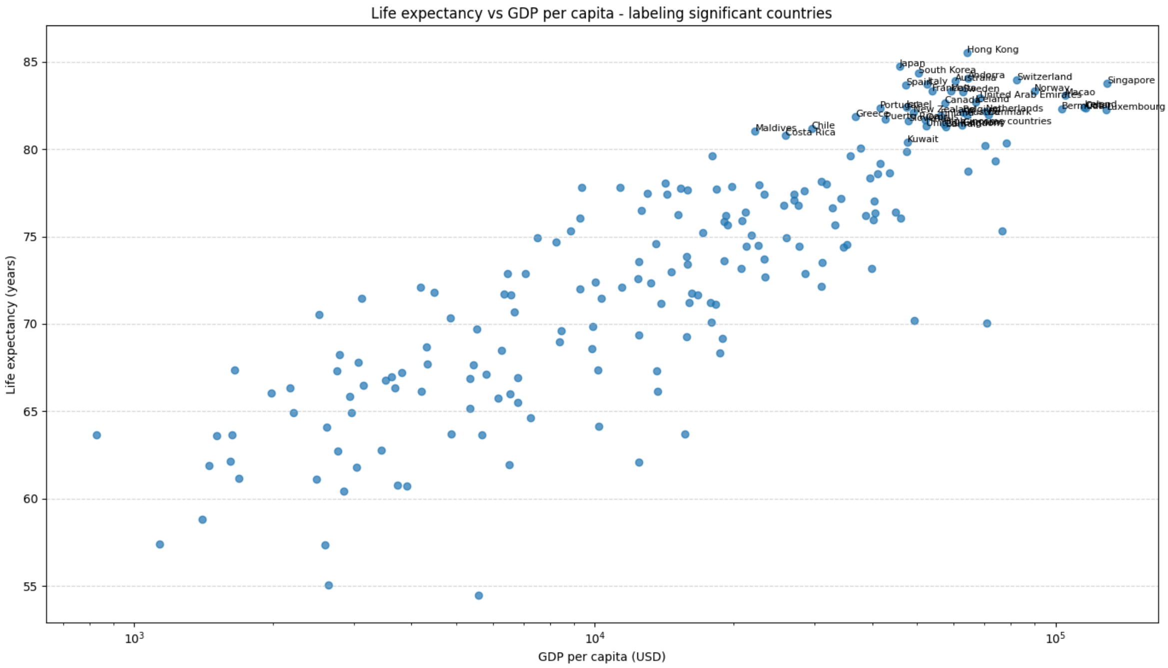
```
                 Entity  Life_expectancy  GDP_per_capita
0             Hong Kong          85.5106       64428.730
1                 Japan          84.7123       45858.660
2           South Korea          84.3288       50414.207
3               Andorra          84.0406       64631.297
4           Switzerland          83.9536       82302.050
5             Australia          83.9228       60461.156
6             Singapore          83.7362      129555.240
7                 Italy          83.7157       52725.746
8                 Spain          83.6703       47341.170
9                France          83.3253       54017.875
10               Norway          83.3078       90085.555
11                Malta          83.2989       59293.297
12               Sweden          83.2624       62845.484
13                Macao          83.0749      104961.920
14 United Arab Emirates          82.9091       68577.540
15              Iceland          82.6911       67177.200
16               Canada          82.6302       57517.440
17              Ireland          82.4118      115505.336
18               Israel          82.4080       47527.152
19                Qatar          82.3680      116159.140
20             Portugal          82.3599       41571.020
21              Bermuda          82.3089      103052.350
22           Luxembourg          82.2294      129021.650
23          Netherlands          82.1576       70673.970
24              Belgium          82.1150       62920.645
25          New Zealand          82.0880       49083.508
26              Austria          81.9559       64394.090
27              Denmark          81.9328       71454.520
28              Finland          81.9102       56246.440
29               Greece          81.8568       36854.504
30          Puerto Rico          81.6902       42738.645
31               Cyprus          81.6480       52190.875
32             Slovenia          81.6034       47868.387
33 High-income countries          81.4353       57349.350
34              Germany          81.3777       62686.540
35       United Kingdom          81.3015       52502.914
36              Bahrain          81.2835       57819.906
37                Chile          81.1667       29563.678
38             Maldives          81.0412       22286.980
39           Costa Rica          80.7995       25979.629
40               Kuwait          80.4047       47777.402
```

41 contries are with so high life expectancy as it is visible in the list above.

Let's see where these contries stand on the scatter plot we showed above.

In [39]:
```
plt.figure(figsize=(14,8))
plt.scatter(merged_df['GDP_per_capita'], merged_df['Life_expectancy'], alpha=0.7
plt.xscale('log')
plt.xlabel(f'GDP per capita (USD)')
plt.ylabel('Life expectancy (years)')
plt.title('Life expectancy vs GDP per capita - labeling significant countries')
plt.grid(axis='y', linestyle='--', alpha=0.5)
```

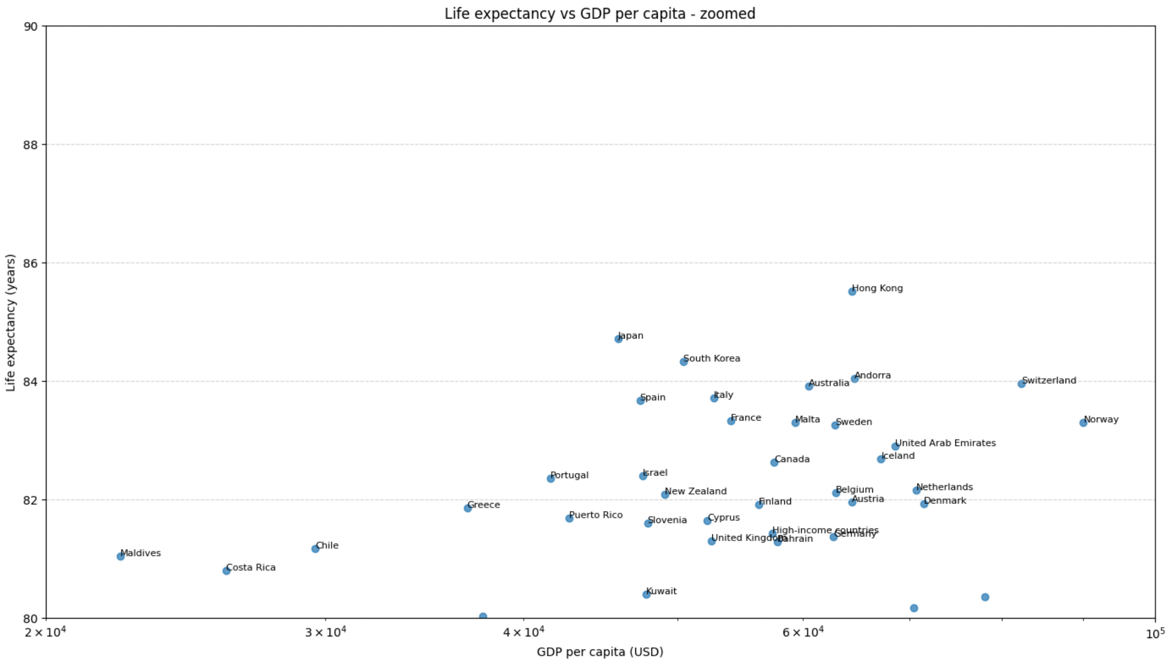
```
significant_countries = pd.concat([
    high_life_expectancy,
]).drop_duplicates('Entity')

for _, r in significant_countries.iterrows():
    plt.annotate(r['Entity'], (r['GDP_per_capita'], r['Life_expectancy']), fonts

plt.tight_layout()
plt.show()
```



The contries with life expectancy higher than one standard deviation above the mean are mostly with very high GDP per capita (they are situated in the upper right corner of the scatter plot)

Let's zoom it in to see where

In [40]:
```
plt.figure(figsize=(14,8))
plt.scatter(merged_df['GDP_per_capita'], merged_df['Life_expectancy'], alpha=0.7
plt.xscale('log')
plt.xlabel(f'GDP per capita (USD)')
plt.ylabel('Life expectancy (years)')
plt.title('Life expectancy vs GDP per capita - zoomed')
plt.grid(axis='y', linestyle='--', alpha=0.5)

significant_countries = pd.concat([
    high_life_expectancy,
]).drop_duplicates('Entity')

for _, r in significant_countries.iterrows():
    plt.annotate(r['Entity'], (r['GDP_per_capita'], r['Life_expectancy']), fonts

plt.xlim(2e4, 1e5)
plt.ylim(80, 90)

plt.tight_layout()
plt.show()
```



c. Which countries have high life expectancy but have low GDP? Motivate how you have chosen to define “high” and “low”

We can define Low and High depending on the situation and the needs of the experiment. It can be defined just above and below the mean value. In the first experiment, we will check for the countries which have GDP per capita in the first quantile, meaning below 25%. For now, we will keep the definition of High life expectancy from the previous task.

In [42]:
```
gdp_quantile_25 = merged_df['GDP_per_capita'].quantile(0.25)
life_expectancy_requirements = life_expectancy_mean + life_expectancy_std
high_life_exp_and_low_gdp = merged_df[(merged_df['Life_expectancy'] > life_expec

high_life_exp_and_low_gdp[['Entity','Life_expectancy','GDP_per_capita']]
```

Out[42]:

Entity	Life_expectancy	GDP_per_capita

As we see, such countries doesn't exist. Let's try with GDP below the mean value.

In [48]:
```
gdp_quantile_50 = merged_df['GDP_per_capita'].mean()
high_life_exp_and_low_gdp = merged_df[(merged_df['Life_expectancy'] > life_expec

high_life_exp_and_low_gdp[['Entity','Life_expectancy','GDP_per_capita']]
```

Out[48]:

	Entity	Life_expectancy	GDP_per_capita
113	Maldives	81.0412	22286.980
42	Costa Rica	80.7995	25979.629

We can see that Maldives and Costa Rica are such countries, most probably because of the good conditions in the countries.

In conclusion, we will decrease the life expectancy threshold to the mean value.

In [50]:
```
gdp_quantile_50 = merged_df['GDP_per_capita'].mean()
high_life_exp_and_low_gdp = merged_df[(merged_df['Life_expectancy'] > life_expec

high_life_exp_and_low_gdp[['Entity','Life_expectancy','GDP_per_capita']]
```

Out[50]:

	Entity	Life_expectancy	GDP_per_capita
190	Ukraine	73.4225	15916.9580
18	Belize	73.5656	12455.2960
174	Suriname	73.6310	19043.7050
52	Dominican Republic	73.7201	23282.4900
143	Paraguay	73.8443	15826.0880
11	Azerbaijan	74.4286	21310.7380
67	Georgia	74.4965	22590.5300
198	Vietnam	74.5883	13545.9350
14	Bangladesh	74.6723	8242.3980
118	Mauritius	74.9263	26060.7320
132	Nicaragua	74.9468	7496.5293
119	Mexico	75.0690	21905.1780
72	Grenada	75.2052	17128.6970
124	Morocco	75.3128	8868.7120
7	Armenia	75.6827	19402.7520
25	Brazil	75.8482	19079.8120
194	Upper-middle-income countries	75.8812	20882.4120
33	Cape Verde	76.0587	9280.9795
15	Barbados	76.1782	19224.4630
2	Algeria	76.2610	15159.3240
180	Thailand	76.4115	21191.4530
184	Tunisia	76.5077	12622.9600
160	Serbia	76.7693	25740.2030
54	Ecuador	77.3924	14343.0190
135	North Macedonia	77.3947	23343.6020
172	Sri Lanka	77.4828	13024.9400
85	Iran	77.6545	15912.0300
39	Colombia	77.7246	18382.9980
144	Peru	77.7401	15327.5820
92	Jordan	77.8145	9381.3545
101	Lebanon	77.8170	11330.2690
23	Bosnia and Herzegovina	77.8504	19803.8460
38	China	77.9529	22687.0760

	Entity	Life_expectancy	GDP_per_capita
96	Kosovo	78.0326	14240.2030
1	Albania	79.6019	17991.0160
42	Costa Rica	80.7995	25979.6290
113	Maldives	81.0412	22286.9800

In this example much more countries exist, however, the requirements are just life expectancy above the mean value and GDP per capita below the mean value which involves countries right in the middle. This last result is not giving correct enough information on the subject.

d. Does every strong economy (normally indicated by GDP) have high life expectancy?

In [54]:
```
top_economy = merged_df['GDP_per_capita'].quantile(0.75)
strong_economy = merged_df[merged_df['GDP_per_capita'] >= top_economy].sort_valu
strong_economy_low_life_expectancy = strong_economy[ strong_economy['Life_expect

strong_economy_low_life_expectancy
```

Out[54]:

	Entity	GDP_per_capita	Life_expectancy
71	Greenland	71037.54	70.0550
76	Guyana	49312.36	70.1802

Although only two countries complete the requirements, this is prooving that the high economy is not ensuring high life expectancy.

e. Related to question d, what would happen if you use GDP per capita as an indicator of strong economy? Explain the results you obtained, and discuss any insights you get from comparing the results of d and e

In [56]:
```
correlation = merged_df[['GDP_per_capita','Life_expectancy']].corr().iloc[0,1]
print(f"Correlation between GDP per capita and life expectancy: {correlation:.3f
```

```
Correlation between GDP per capita and life expectancy: 0.744
```

In conclusion, we can see that the correlation is almost 0.75 which is in the highest 25%, however it is not enough to conclude that high GDP per capita results in a strong economy.